Project Design Phase

Project Title :ComicCraft – AI Comic Story Creator Using Gemini Models

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1. Introduction

Requirement Analysis is the phase in which the needs, functions, resources, and expectations of the ComicCraft system are identified and documented. The main purpose is to clearly define what the system should do and what users require from it.

2. Project Requirements

ComicCraft should provide an easy-to-use platform that allows users to enter a story idea or prompt and receive AI-generated comic content using Gemini models.

3. Functional Requirements

• User

1. Introduction

The Project Design Phase defines the structure, architecture, modules, user interface, data flow, and interaction of the ComicCraft system. The design is prepared based on the requirements identified in the Requirement Analysis Phase.

2. System Architecture

ComicCraft can be designed using a layered architecture consisting of the User Interface Layer, Application/Backend Layer, Gemini AI Service Layer, and Data Storage Layer. The user submits a prompt through the interface. The application sends the prompt to the Gemini model, receives the generated content, processes it, and displays the result to the user.

3. Main System Modules

- User Interface Module – Provides screens for entering prompts and viewing generated comic content.
- Prompt Processing Module – Validates and prepares user input for AI processing.
- Gemini AI Integration Module – Connects the application with Gemini models and handles AI requests and responses.
- Story Generation Module – Produces the storyline, characters, dialogues, and scene descriptions.
- Editing Module – Allows users to modify and regenerate generated content.
- Storage Module – Stores user projects and generated content when storage is included.
- Export/Preview Module – Displays the comic in an organized format and supports export where implemented.

4. User Interface Design

The interface should be simple and user-friendly. The main screen can contain a prompt input area, generation button, output area, editing controls, regeneration option, and save/export options. Generated content should be organized into story sections or comic panels for easy understanding.

5. Data Flow Design

1. User enters a story idea or prompt.
2. The system validates the input.
3. The prompt is sent to the Gemini model through the application.

4. Gemini processes the prompt and generates comic content.

5. The application receives and organizes the response.

6. The generated story, characters, dialogues, and scenes are displayed to the user.

7. The user can edit, regenerate, save, or export the content.

6. Database / Storage Design

If persistent storage is implemented, the system can maintain user information, project details, prompts, generated stories, characters, scenes, and saved versions. A suitable database or cloud storage service can be selected based on the final implementation.

7. Security Design

API keys and sensitive configuration information should be protected and should not be exposed in the client-side interface. User data should be handled securely, and access to stored projects should be controlled appropriately.

8. Design Considerations

- Simple and responsive interface.
- Modular system structure.
- Easy integration with Gemini models.
- Ability to extend the system with image generation and additional comic styles.
- Error handling for invalid prompts, API failures, and unavailable services.

9. Expected Outcome

The Project Design Phase produces a clear technical blueprint for ComicCraft. It defines how users interact with the system, how prompts are processed, how Gemini models are integrated, and how generated comic content is displayed and stored.